

3.22

0xC0000000 in binary: 0000 1100 0000 0000 0000 0000 0000 0000

Sign = 0 (positive)

Exponent bits = 00011000 = 24, unbiased = $24 - 127 = -103$

Mantissa = all zeros, so significand = 1.0 (hidden bit)

$$\text{Value} = +1.0 \times 2^{(-103)} = 9.8607613 \times 10^{(-32)}$$

3.24

$63 = 111111$, $0.25 = .01$, so $63.25 = 111111.01$

Normalized: 1.1111101×2^5

Sign = 0

$$\text{Exponent} = 5 + 1023 = 1028 = 10000000100$$

Mantissa = 1111101 then 45 zeros

Answer: 0 10000000100 1111101000

3.25

IBM single precision: 1 sign, 7 exponent (excess-64), 24 mantissa, base 16

63.25 decimal = 3F.4 hex = 0.3F4 x 16²

Sign = 0

$$\text{Exponent} = 2 + 64 = 66 = 1000010$$

Mantissa: 0.3F4 hex = 0011 1111 0100 0000 0000 0000

Answer: 0 1000010 0011 1111 0100 0000 0000 0000 = 0x423F4000

3.27

Modified half precision: 1 sign, 5 exponent (excess-16), 10 mantissa, hidden 1

$$-1.5625 \times 10^{(-1)} = -0.15625$$

$$0.15625 \text{ in binary: } 0.00101 = 1.01 \times 2^{(-3)}$$

Sign = 1 (negative)

$$\text{Exponent} = -3 + 16 = 13 = 01101$$

$$\text{Mantissa} = 0100000000$$

$$\text{Answer: } 1\ 01101\ 0100000000 = 0xB500$$

3.29

$$A = 2.6125 \times 10^1 = 26.125 = 11010.001 = 1.1010001 \times 2^4$$

$$\text{Stored A: sign}=0, \text{ exp}=4+16=20=10100, \text{ mantissa}=1010001000$$

$$A = 0\ 10100\ 1010001000$$

$$B = 4.150390625 \times 10^{(-1)} = 0.4150390625 = 425/1024 = 0.0110101001 = 1.10101001 \times 2^{(-2)}$$

$$\text{Stored B: sign}=0, \text{ exp}=-2+16=14=01110, \text{ mantissa}=1010100100$$

$$B = 0\ 01110\ 1010100100$$

Align exponents: difference = $4 - (-2) = 6$, shift B right by 6

B significand: $1.1010100100 \gg 6 = 0.0000011010$ with bits shifted out: 100100

Guard=1, Round=0, Sticky=OR(0,1,0,0)=1, so GRS = 1,0,1

Add:

$$1.1010001000$$

$$+ 0.0000011010$$

= 1.1010100010

GRS still 1,0,1

Round: GRS=101 means remainder > 0.5 ULP, so round up

$1.1010100010 + 0.0000000001 = 1.1010100011$

Final: 0 10100 1010100011

Decimal value: $1.1010100011 \times 2^4 = 26.546875$

Exact sum = 26.5400390625